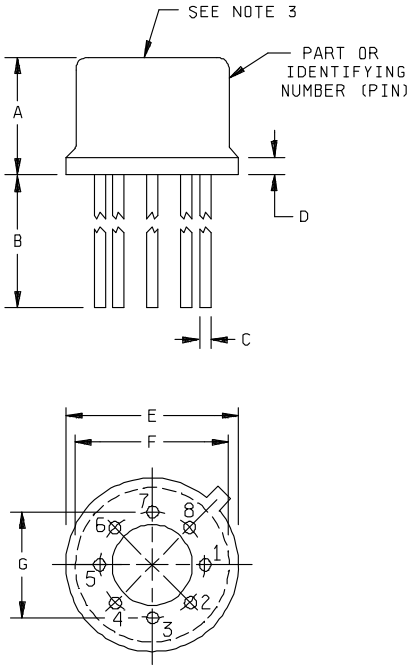
INCH-POUND
MIL-PRF-55310/13J
9 August 2010
SUPERSEDING
MIL-PRF-55310/13H
11 June 2009

PERFORMANCE SPECIFICATION SHEET

# OSCILLATOR, CRYSTAL CONTROLLED, TYPE 1 (CRYSTAL OSCILLATOR (XO)), 300 Hz THROUGH 10 MHz, HERMETIC SEAL, SQUARE WAVE, CMOS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-55310.




| Pin number | Function |
|---|---|
| 1 | NC |
| 2 | NC |
| 3 | NC |
| 4 | B - (GND) |
| 5 | OUTPUT |
| 6 | NC |
| 7 | NC |
| 8 | B+ |

| Ltr | Inches | | mm | |
|---|---|---|---|---|
| | Min | Max | Min | Max |
| A | --- | .350 | --- | 8.89 |
| B | .480 | --- | 12.19 | --- |
| C | .016 | .020 | 0.41 | 0.51 |
| D | --- | .040 | --- | 1.02 |
| E | --- | .610 | --- | 15.49 |
| F | --- | .502 | --- | 12.75 |
| G | .277 | .287 | 7.04 | 7.29 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3 Standard marking, except omit Part or Identifying Number (PIN) this location.
4. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.
5. Case to be connected to base pin 4 to permit shielding of the oscillator.
6. All pins with NC function may be connected internally and are not to be used as external tie points or connections.
7. Pin numbers shown on package are for reference only and need not appear on part.

FIGURE 1. Dimensions and configuration.



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REQUIREMENTS:

Interface and physical dimensions: See figure 1.

Mounting: See figure 1.

Terminals: See figure 1.

Seal: Hermetic in accordance with MIL-PRF-55310, maximum leakage rate 5 x $10^{-8}$ atm cc/s.

Weight: 0.175 ounce, maximum.

Oscillator: Class 2 or any class 1 or class 3 oscillator meeting all class 2 requirements and verification tests specified herein and in MIL-PRF-55310.

Calibration: Manufacturer calibrated.

Screening: In accordance with MIL-PRF-55310, product level B or S, as applicable.

Temperature:

Operating: See table I.

Storage: -62°C to +125°C.

Oscillator load: 10 kilohms ±5 percent shunted by a 15 pF ±5 percent capacitor for a CMOS compatible square wave output.

Output waveform: Symmetrical square wave.

Supply voltage: See table I.

Input current: At designated supply voltage (see table I).

Output frequency: Frequency as designated at time of acquisition (see table I).

Output voltage:

Logic 1: See table I.

Logic 0: See table I.

Rise and fall times: See table I.

Duty cycle: See table I.

Initial accuracy at reference temperature: ±25 ppm at +23°C ±1°C up to 30 days after shipment.

Initial frequency-temperature accuracy (one-half temperature cycle): Verification applicable. 1/



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TABLE I. Dash numbers and operating characteristics.

| Dash number | Output frequency range | Supply voltage | Input current (max) | Output voltage | | Rise and fall times (max) | Duty cycle | Frequency temperature tolerance (ppm) | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Logic: 1 (min) | Logic: 0 (max) | | | -55°C to +125°C | -55°C to +105°C | -20°C to +70°C |
| | | | | | | | | A | B | C |
| | | V dc | mA | | | ns | percent | | | |
| 01 | 300 Hz to 6 MHz | +15 | 40 | 13 | 2 | 30 | 45 to 55 | ±100 | ±70 | ±55 |
| 02 | 6 MHz to 10 MHz | +15 | 40 | 13 | 2 | 30 | 45 to 55 | ±100 | ±70 | ±55 |
| 03 | 300 Hz to 6 MHz | +12 | 30 | 10 | 1 | 60 | 45 to 55 | ±100 | ±70 | ±55 |
| 04 | 6 MHz to 10 MHz | +12 | 30 | 10 | 1 | 60 | 45 to 55 | ±100 | ±70 | ±55 |
| 05 | 300 Hz to 6 MHz | +10 | 20 | 9.5 | 0.5 | 60 | 45 to 55 | ±100 | ±70 | ±55 |
| 06 | 6 MHz to 10 MHz | +10 | 20 | 9.5 | 0.5 | 60 | 45 to 55 | ±100 | ±70 | ±55 |

Frequency-temperature tolerance (one-half temperature cycle, referenced to frequency measured at +23°C ±1°C, immediately prior to starting of the test): See table I. Measurements taken at ten equally spaced increments over the specified operating temperature range. 1/

Frequency-voltage tolerance: ±1 ppm maximum for a ±10 percent change in supply voltage. Measurements taken at reference temperature and operating temperature range end points.

Frequency aging: Measurements shall be taken at +70°C ±0.2° C at intervals of not more than every 72 hours for a minimum of 30 days.

±0.5 ppm per 30 days, maximum.

±2 ppm per 90 days, maximum. 2/

±10 ppm per year, maximum. 2/

Terminal strength: Method 211 of MIL-STD-202, test condition C.

Applied force: 2 pounds each terminal for 10 seconds.

Bends: Five at 45 degrees each.

Frequency-environmental tolerance: Not applicable.

Vibration, sinusoidal: In accordance with MIL-PRF-55310 and method 204 of MIL-STD-202.

Nonoperating: Test condition D.

Operating: Not required.

1/ For the purpose of transitioning this device to MIL-PRF-55310, 'Frequency stability versus temperature' has been renamed 'Frequency-temperature tolerance'. The verification requirements of 'initial frequency-temperature accuracy (one-half temperature cycle)' shall apply except that frequency measurements shall be referenced to the frequency measured at +23°C ±1°C ($f_{ref}$) instead of to the nominal frequency ($f_{nom}$).

2/ This is a performance requirement of the oscillator. Support data shall be presented to the qualifying activity showing that this requirement shall be met.



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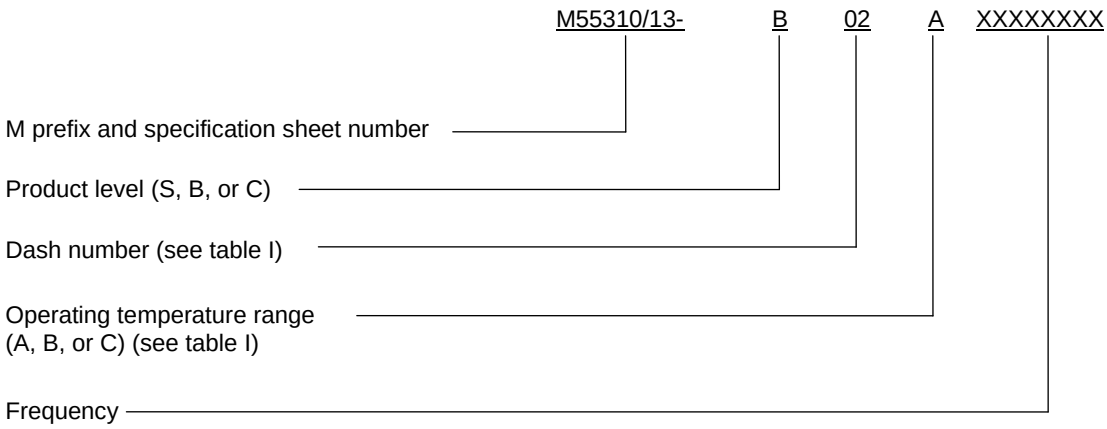
Ambient pressure:

Nonoperating: In accordance with MIL-PRF-55310.

Operating: Method 105 of MIL-STD-202, test condition C.

Part or Identifying Number (PIN): Consists of "M" prefix followed by specification sheet number, a dash and coded alphas, and numeric number. See example:

EXAMPLE




Reference documents. In addition to MIL-PRF-55310, this document references the following:

MIL-STD-202

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Custodians:
Army - CR
Navy - EC
Air Force - 99
DLA - CC

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5955-2010-008)

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
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